

MSE435: AI Economics and Infrastructure

Stanford Frontier AI Learning System

Baseline: October 6, 2026. Print companion to the HTML course site:
course map, prerequisites, notation and shapes, glossary,
cheatsheet, crash course.

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Course map

The nine-week materials list anchors six teaching units. Session titles and guests come from the inspected materials page (SRC-00, fetched 2026-10-07). Claim classes: OFFICIAL-SCHEDULE for topics named on the page, REQUESTED-BRANCH for requested prerequisite and implementation depth. All leaves: PLANNED / SOURCE ATTRIBUTION PENDING.

Week	Session (official)	Guest(s)	Unit(s)	Claim class
1	Intro to the Economics of Generative AI	course staff	U01	OFFICIAL-SCHEDULE
2	Silicon , The GPU Economy	Brad Gerstner (Altimeter), Jonathan Ross (Groq)	U02	OFFICIAL-SCHEDULE
3	Infrastructure, Building AI Factories at Gigawatt Scale	Chase Lochmiller (Crusoe)	U02	OFFICIAL-SCHEDULE

Week	Session (official)	Guest(s)	Unit(s)	Claim class
4	Infrastructure, Enterprise AI, and Service As a Software	Ali Ghodsi (Databricks)	U03	OFFICIAL-SCHEDULE
5	Infrastructure and Capstone Case	Sachin Katti (OpenAI)	U03 (infra + token supply/demand), U06 (capstone bridge)	OFFICIAL-SCHEDULE
6	Intelligence, Unlocking Enterprise Internal Knowledge	Yash Patil (Applied Intelligence)	U04 (built 2026-10-07)	OFFICIAL-SCHEDULE

Week	Session (official)	Guest(s)	Unit(s)	Claim class
7	Inference, Building the Inference Cloud	Tuhin Srivastava (Baseten)	U04 (built 2026-10-07)	OFFICIAL-SCHEDULE
8	Applications, AI in Life Sciences	Eric Kauderer-Abrams (Anthropic), Josh Meier (Chai Discovery)	U05 (built 2026-10-07)	OFFICIAL-SCHEDULE
9	Capstone defense (no separate materials entry)	n/a	U06 (built 2026-10-07)	REQUESTED (course framing)

Unit summary

- U01, AI economics and the full value chain. Week 1. Bridges P01, P23. This build.
- U02, Silicon, power, and data centers. Weeks 2-3. Bridges P15, P23. This build.
- U03, Enterprise infrastructure and token markets. Weeks 4-5. Bridges P16, P23, P24. This build.
- U04, Enterprise knowledge and inference cloud. Weeks 6-7. Bridges P19, P23, P24. Built 2026-10-07 (second builder).
- U05, Coding/application and life-science economics. Week 8. Bridges P21, P22, P23, P24. Built 2026-10-07 (second builder).
- U06, Capstone evidence and stakeholder defense. Weeks 5 (case launch) and 9 (defense). Bridges P07, P22, P23, P24. Built 2026-10-07 (second builder).

Prerequisite bridges

Shared prerequisite modules live at `../shared/prerequisites/` and are linked, not rebuilt. Local remediation for U01-U06 sits in `prerequisites.md`.

Prerequisites

Shared prerequisite modules are linked, not rebuilt. Each unit below links its bridges and carries local remediation: the smallest self-contained repair that lets a learner continue without leaving this file.

Shared bridges (linked)

- P01 , Numeracy, algebra, and notation:

`../shared/prerequisites/p01_numeracy.md`

- P15 , Hardware and computer architecture:

`../shared/prerequisites/p15_hardware.md`

- P16 , Distributed systems and networking:

`../shared/prerequisites/p16_distributed.md`

- P23 , Economics and decision analysis:

`../shared/prerequisites/p23_economics.md`

- P24 , Production ML and stakeholder foundations:

`../shared/prerequisites/p24_production_ml.md`

U01 , bridges P01, P23

Local remediation U01-R1 , numbers with units. Every quantity in U01 carries a unit: dollars, tokens, hours, or dollars per unit. Check the unit of each term before adding two numbers. If the units do not match, the addition is wrong even when the arithmetic is right. Example: 100 dollars of fixed cost + 5 dollars per token of variable cost is not 105 dollars. It is $100 + 5x$, where x counts tokens.

Local remediation U01-R2 , percentages and ratios. A 10 percent price cut on a 20 dollar base gives 18 dollars, computed as $20 \times (1 - 0.10)$. A ratio has no unit: cost divided by revenue is a pure number. A percentage point is not a percent: a margin that moves from 20 percent to 25 percent rose 5 percentage points, which is a 25 percent relative rise.

Local remediation U01-R3 , supply and demand reading. Read P23 sections on supply/demand, fixed/variable costs, capital/operating expense, and unit economics. The U01 lesson re-derives every curve it uses, so P23 is a reference, not a gate.

U02 , bridges P15, P23

Local remediation U02-R1 , power and energy. Power is a rate, measured in watts (joules per second). Energy is an amount, measured in watt-hours or joules. A GPU that draws 700 W for one hour uses 0.7 kWh of energy. Cost scales with energy, capacity scales with power. Confusing the two breaks every data-center calculation in U02.

Local remediation U02-R2 , FLOPs versus FLOP/s. FLOPs count operations (an amount). FLOP/s counts operations per second (a rate). A training run needs FLOPs. A cluster delivers FLOP/s. Time equals FLOPs divided by FLOP/s, adjusted for utilization below 1.

Local remediation U02-R3 , straight-line depreciation. An asset that costs C dollars and lasts T years loses C / T dollars of book value per year. The U02 lesson re-derives the per-GPU-hour capital charge from this rule.

U03 , bridges P16, P23, P24

Local remediation U03-R1 , tokens as the unit of account. One million tokens is the billing block in the U03 toy market. Price quotes in dollars per 1M tokens convert to dollars per token by dividing by 1,000,000. Keep the block unit explicit in every calculation.

Local remediation U03-R2 , contracts as cash-flow series. A contract that pays X dollars per year for T years has present value sum over t of $X / (1 + r)^t$, with discount rate r. The U03 lesson re-derives this for the long-contract concept.

Local remediation U03-R3 , latency versus throughput. Read P16 on latency versus throughput before U03-C06 and U03-C07. Batching raises throughput and usually raises latency. The lesson shows the trade with numbers.

U04 , bridges P19, P23, P24

Local remediation U04-R1 , hit rate arithmetic. The hit rate h is a share between 0 and 1. Effective price = $h \times \text{cheap} + (1 - h) \times \text{dear}$. At $h = 0.6$, cheap 0.25, dear 4.00: $0.15 + 1.60 = 1.75$. Measure h on your own traffic. Vendor slides overstate it.

Local remediation U04-R2 , break-even division. $Q^* = F / (p_a - p_h)$. Fixed cost in dollars per month divided by a price difference in dollars per block gives blocks per month. If $p_a - p_h$ is negative, hosting never wins: check the sign before dividing.

Local remediation U04-R3 , retry compounding. Success with r retries = $1 - (1 - s)^{(r+1)}$, a power, not a product. At $s = 0.6$, $r = 1$: $1 - 0.16 = 0.84$.

U05 , bridges P21, P22, P23, P24

Local remediation U05-R1 , finite horizons. A 36-month contract gives at most 36 months of retained life, not the infinite-horizon $1/\text{churn}$. At 3 percent monthly churn the finite life is 22.2 months. Use the contract length.

Local remediation U05-R2 , funnel multiplication. Funnel cost = sum over stages of survivors x cost per survivor. Doubling survivors doubles every downstream stage's cost. Add, do not average.

Local remediation U05-R3 , expected loss. Expected loss = probability x loss. Buy protection when its price is below the expected loss with margin.

U06 , bridges P07, P22, P23, P24

Local remediation U06-R1 , TCO addition. $\text{TCO} = \text{build} + 3 \times (\text{run} + \text{labor}) + \text{risk reserve}$. Build is paid once, run and labor recur, the reserve is a cushion. Never present build as the total.

Local remediation U06-R2 , ramped present value. Savings in year t are $a_t \times S / (1 + r)^t$ with the adoption ramp a_t , not $S / (1 + r)^t$. The ramp $[0.20, 0.50, 0.80]$ cuts the naive PV by more than half.

Local remediation U06-R3 , pre-registration. Fix the threshold Y , the duration D , and the instrument before the pilot starts. Invest iff the interval's lower bound L exceeds Y . Moving Y or D after seeing data voids the decision.

Answer closed-book. Score with `keys/diagnostic_key.md`. Any score below 8 means: read the matching shared bridge first, then the local remediation above.

1. A server costs 12,000 dollars and draws 1,200 W. Electricity

costs 0.10 dollars per kWh. What is the energy cost of running it for 30 days at full draw? (U02-R1)

1. Fixed cost 50,000 dollars per month, variable cost 3 dollars per

unit, price 8 dollars per unit. What monthly volume breaks even? (U01-R1)

1. A price falls from 40 dollars to 30 dollars. State the change in

percentage points is wrong here, why, and give the percent change. (U01-R2)

1. A workload needs 10^{18} FLOPs. A cluster delivers 10^{15} FLOP/s at

40 percent utilization. How many seconds does the run take? (U02-R2)

1. An asset costs 90,000 dollars and lasts 3 years. What is the straight-line depreciation per year? (U02-R3)
1. A token price is 4 dollars per 1M tokens. A request uses 2,000 tokens. What does the request cost? (U03-R1)
1. A contract pays 100,000 dollars at the end of each of 2 years. Discount rate 10 percent. What is the present value? (U03-R2)
1. Demand is $Q = 100 - 2P$. Supply is $Q = 20 + 3P$. What are the equilibrium price and quantity? (U01)
1. Raising batch size from 1 to 8 doubles throughput but triples latency. A user needs latency under 2 seconds and the 8-batch latency is 3 seconds. Which constraint binds? (U03-R3)
1. Fixed cost 0, variable cost 5 dollars per unit, price 5 dollars per unit. What is the margin per unit, and what does a zero margin imply for scaling losses? (U01-R1)

Notation and shapes

Every symbol used in U01-U06 lessons, with units. No symbol is used before its row appears in a lesson.

Market and firm economics

Symbol	Meaning	Unit
P	price	dollars per unit
Q	quantity	units (tokens, GPU-hours)
Qd	quantity demanded	units
Qs	quantity supplied	units

Symbol	Meaning	Unit
P*	equilibrium price	dollars per unit
Q*	equilibrium quantity	units
MC	marginal cost	dollars per unit
FC	fixed cost	dollars per period

Symbol	Meaning	Unit
VC	variable cost	dollars per period
v	variable cost per unit	dollars per unit
TC	total cost = $FC + v \times Q$	dollars per period
ATC	average total cost = TC / Q	dollars per unit

Symbol	Meaning	Unit
AVC	average variable cost	dollars per unit
R	revenue = $P \times Q$	dollars per period
pi	profit = $R - TC$	dollars per period
m	unit margin = $P - v$	dollars per unit

Symbol	Meaning	Unit
μ	margin ratio = $(P - ATC) / P$	pure number
capex	capital expenditure	dollars (one-time)
opex	operating expenditure	dollars per period
r	discount rate	per year

Symbol	Meaning	Unit
T	horizon	years
NPV	net present value	dollars
IRR	internal rate of return	per year
Ep	price elasticity of demand	pure number

Compute and infrastructure

Symbol	Meaning	Unit
W	power draw	watts
E	energy	watt-hours
PUE	power usage effectiveness	pure number, ≥ 1
U	utilization	pure number, 0 to 1

Symbol	Meaning	Unit
F	FLOPs required	operations
S	throughput	FLOP/s
B	memory bandwidth	bytes per second
C_gpu	GPU-hours used	hours

Symbol	Meaning	Unit
c_h	cost per GPU-hour	dollars per hour
MW	megawatt, 10^6 W	watts
GW	gigawatt, 10^9 W	watts
TCO	total cost of ownership	dollars

Token markets

Symbol	Meaning	Unit
p_tok	token price	dollars per 1M tokens
D_tok	token demand	1M-token blocks per period
S_tok	token supply (serving capacity)	1M-token blocks per period
L	latency	seconds

Symbol	Meaning	Unit
X	throughput	tokens per second
B_sz	batch size	requests
c_task	cost per successful task	dollars
h	cache hit rate	pure number, 0 to 1

Symbol	Meaning	Unit
c_c	cache-serve price	dollars per 1M tokens
c_m	model-call (miss) price	dollars per 1M tokens
p_a	API price	dollars per 1M tokens
p_h	hosted price	dollars per 1M tokens

Symbol	Meaning	Unit
F	fixed hosting cost	dollars per month
s	task success rate	pure number, 0 to 1
e	escalation (human review) rate	pure number, 0 to 1
r_trials	retries after the first attempt	count

Symbol	Meaning	Unit
a0, a1	answer accuracy before/after access	pure number, 0 to 1
q	questions or tasks per period	count per period
v_qual	value per quality point	dollars per point per period
w_q	value weight per accuracy point	dollars per point

Symbol	Meaning	Unit
B	build cost	dollars (one-time)
R	run cost	dollars per year
Lb	labor cost	dollars per year
K	risk reserve	dollars

Symbol	Meaning	Unit
Y	pilot investment threshold	pure number
D	pilot duration	weeks
L	interval lower bound	pure number
a _t	adoption ramp in year t	pure number, 0 to 1

Conventions

- Dollars are nominal at the 2026-10-06 baseline unless labeled real.
- A subscript marks the unit of account: p_{tok} is not the same object as P (per-unit price of a different good).
- Toy numbers in lessons and figures are computed locally and labeled TOY. "Not in source" marks any value no inspected source provides.

Glossary

One meaning per term. The lesson uses the term exactly this way.

- adoption , share of potential users or firms that actually use a technology in a period.
- average total cost , total cost divided by quantity, TC / Q .
- batching , grouping multiple inference requests so one model pass serves them together.
- bottleneck , the stage or resource that limits total throughput of a system.
- break-even , the volume at which revenue equals total cost.
- build/lease , the choice between owning a data center and renting capacity from a provider.
- capex , capital expenditure: one-time spend on long-lived assets.
- capacity , the maximum quantity a system can supply per period.
- complements , goods used together, so a price fall in one raises demand for the other.
- consumer AI , AI products sold to individuals, priced per user or free with ads.
- contribution margin , price minus variable cost per unit, $P - v$.
- cooling , removal of waste heat from chips, by air or liquid.
- custom accelerator , a chip designed for one workload class, such as an ASIC or TPU.
- data governance , rules for who may access data, where it lives, and how long it is kept.

- depreciation , accounting spread of an asset's cost over its useful life.
- discount rate , the rate that converts future dollars to present dollars.
- enterprise AI , AI products sold to firms, priced per seat, per task, or by contract.
- equilibrium , the price and quantity where quantity demanded equals quantity supplied.
- fixed cost , cost that does not change with quantity in the period.
- forecast , a forward-looking claim about an unknown quantity, always uncertain.
- gigawatt scale , data-center builds at or above 1 GW of power, roughly a large power plant.
- GPU economy , the market for GPU compute: supply, price, and allocation of accelerators.
- inference , running a trained model to produce outputs.
- interconnect , the network links between chips and servers that move activations and gradients.
- long contract , a multi-year commit to buy compute or tokens at set terms.
- marginal cost , the cost of producing one more unit.
- memory bottleneck , the state where memory bandwidth, not compute,

limits speed.

- NPV , net present value: present value of cash inflows minus outflows.
- opex , operating expenditure: recurring spend to run the business.
- open/closed stack , whether model weights, APIs, and infra are freely usable or vendor-controlled.
- productivity , output per unit of input, such as per worker hour.
- PUE , power usage effectiveness: total facility power divided by IT equipment power.
- scenario analysis , recomputation of outcomes under a small set of named alternative assumptions.
- sensitivity , how much an outcome moves when one input moves.
- service-as-software , the thesis that AI turns labor-like services into software-priced products.
- substitutes , goods that replace each other, so a price fall in one lowers demand for the other.
- supply concentration , the share of supply held by few producers.
- token , the billing unit of LLM input and output text.
- token demand , quantity of tokens buyers want at a price.
- tornado chart , a bar chart that ranks inputs by their effect on an outcome.
- training , fitting model weights on data, a one-time large compute cost.
- uncertainty , the range of plausible outcomes around a forecast.

- unit price , price per billing block, such as dollars per 1M

tokens.

- utilization , used capacity divided by total capacity.

- value capture , the share of total created value that a firm

keeps as profit.

- value chain , the ordered layers from raw input to final product,

here chips to applications.

- variable cost , cost that rises with quantity in the period.

- vendor lock-in , the cost of leaving a vendor once systems depend

on it.

- cache hit rate , the share of requests served from cache, h.

- clean room , an isolated environment where proprietary data is

processed without leaking to shared models.

- cost per success , expected cost of one successful task,

$(r+1) \times c / \text{success}$.

- data flywheel , the loop where usage corrects data, corrections

raise quality, and quality raises usage.

- escalation rate , the share of AI outputs sent to human review,

e.

- falsifiable , stated so that a named measurement can prove it

wrong.

- fixed cost floor , the minimum cost set by infrastructure that

does not scale with volume.

- guardrail , a pre-registered limit that blocks launch or forces rollback when breached.
- hit rate , see cache hit rate.
- human escalation , see escalation rate.
- kill criteria , the named conditions under which a pilot or project stops.
- pre-register , to fix the decision rule, threshold, and duration before seeing the data.
- ramp , the assumed adoption path over time, such as [0.20, 0.50, 0.80].
- review labor , human time spent checking AI output, priced per task as $\text{tasks} \times e \times \text{minutes}/60 \times \text{wage}$.
- risk reserve , the budget cushion K added to TCO for surprises.
- rollback , restoring the previous working system, here required within 15 minutes.
- shadow mode , running the new system on live traffic without affecting users.
- success rate , the share of tasks completed correctly, s .
- switching cost , the one-time cost of moving from one vendor or design to another.
- TCO , total cost of ownership: $\text{build} + 3 \times (\text{run} + \text{labor}) + \text{risk reserve}$.
- trust boundary , the line across which data may move only under contract.
- verification gap , the cost difference between generating an output and checking it.

Cheatsheet

One page per section. Every formula, every decision rule, every toy constant. Numbers are TOY unless marked.

Master formulas

Name	Formula
Market clearing	$100 - 2P = 20 + 3P$, $P^* = 16$, $Q^* = 68$
Average cost	$F/Q + v$
Unit inference cost	$C_{\text{train}}/V + c_{\text{inf}}$
Servable tokens/day	$\text{GPUs} \times \text{tok/s} \times U \times 86,400$

Name	Formula
PUE	total power / IT power
Depreciation	purchase x 0.6 ^t
Access value	q x (a1 - a0) x v
Flywheel month t	q x corr x gain x decay ^(D-t)

Name	Formula
Hosting break-even	$Q^* = F / (p_a - p_h)$
Cached effective price	$h \times c_c + (1 - h) \times c_m$
Latency tradeoff net	$V \times s \times dL - \text{saving}$
Review labor	$\text{tasks} \times e \times \text{min}/60 \times \text{wage}$

Name	Formula
Cost per success	$(r+1) \times c / (1 - (1-s)^{(r+1)})$
Blended margin	$\text{sum share}_i \times (\text{price}_i - \text{cost}_i)$
Amdahl	$1 / ((1-p) + p/s)$
Funnel total	$\text{sum } n_i \times c_i$

Name	Formula
Savings PV (ramped)	$\sum a_t \times S / (1+r)^t$
TCO 3-year	$B + 3(R + L) + K$
Pilot rule	invest iff $L > Y$
Pilot sample size	$n = (1.96/\text{half-width})^2 \times p(1-p)$

Decision rules

Situation	Rule
Falling token prices	Never sign take-or-pay beyond the visible horizon
Host vs API	Host iff $\text{volume} > F / (p_a - p_h)$
Cache	Build iff $h \times \text{saving} \times \text{volume} > \text{infra}$
Batching	Batch iff $V \times s \times dL < \text{saving}$

Situation	Rule
License vs build	License iff vendor TCO < build TCO over 3 years
Pilot	Pre-register Y, D, instrument. Invest iff $L > Y$
Pilot breach	Kill on two consecutive guardrail breaches
Rollback	15-minute button, game-day tested

Situation	Rule
Quality gate	$a1 \geq 0.75$ binary
Contract	Price-cap or 90-day termination for convenience
Build vs lease	Build base load, lease peaks
Retraining	Add $n_runs \times C_train/V$ when cadence is quarterly

Toy constants (do not quote)

Constant	Value
Token market	$P^* = 16, Q^* = 68$
API price	4.00 per 1M
Hosted price	1.50 per 1M
Cache serve	0.25 per 1M

Constant	Value
Fixed hosting cost	30k per month
Break-even	12B tokens per month
Training cost	5M
Lifetime volume	500B tokens

Constant	Value
GPU throughput	50 tok/s, U = 0.85
Power per GPU	0.7 kW
Cooling overhead	0.3 W per W
PUE target	≤ 1.3

Constant	Value
Depreciation	40 pct per year
Access quality lift	$a_1 - a_0 = 0.37$
Flywheel decay	0.9 per month
Review wage	40 per hour, 10 min per task

Constant	Value
Retry success	$s = 0.6, r = 1, \text{cost } 1.19$
Savings ramp (HYPOTHETICAL)	$[0.20, 0.50, 0.80], r = 10 \text{ pct}$
Savings PV (HYPOTHETICAL)	767.9k

Constant	Value
Lean TCO (HYPOTHETICAL)	680k, net +87.9k
Hosted TCO (HYPOTHETICAL)	1,850k, build share 21.6 pct
Pilot	5,000 cases, 4 weeks, Y = 0.15

Watch-outs

- The pilot average (61.50) is not the steady-state rate. Never quote pilot cost as the unit price.
- Analytic hit-rate models overstate small caches by ~12 percent (capstone 1, H1 FAIL). Measure h.
- Doubling cache 500 to 1,000 gains ~5 points: diminishing, not linear.
- Moving Y or D after the pilot voids the decision.
- Build-only business cases understate cost 3-4x.
- The softest inputs: adoption ramp, success rate, review labor. Instrument all three before investing.
- Speaker forecasts are [SPEAKER CLAIM, EVIDENCE PENDING]. Never build on an unlabeled claim.

Crash course

All six units, every concept, condensed to the one-line definition, the key formula, and the toy number that proves it. Numbers are TOY unless marked observed. Details live in lessons/unit_*.md.

U01 , AI economics and the full value chain

- C01 supply/demand. Price clears where quantity supplied equals quantity demanded. Toy: $Q_d = 100 - 2P$, $Q_s = 20 + 3P$ gives $P^* = 16$, $Q^* = 68$.
- C02 complements/substitutes. Cheaper complements raise demand. Cheaper substitutes steal it. Model: Q_d shifts with the substitute price.
- C03 fixed/variable costs. Fixed costs are paid once. Variable costs scale with output. Average cost = $F/Q + v$.
- C04 capex/opex. Capex buys the asset, opex rents the flow. Depreciation turns capex into a yearly cost.
- C05 value capture. Value created is not value kept. Capture goes to the scarcest layer: margin = price minus cost at the bottleneck.
- C06 layers from chips to applications. Chips, infrastructure, models, applications. Each layer taxes the next.
- C07 consumer versus enterprise. Consumers buy delight, enterprises buy ROI. Enterprise sales cycles are long but contracts are sticky.
- C08 productivity versus adoption. Productivity gains need adoption first. Output = productivity x adoption.
- C09 forecasts versus observations. Forecasts are guesses with error bars. Trust observations, derate forecasts.
- C10 two-year comparison. Compare the same metric at t and $t + 2$ years. Growth compounds, costs decay.
- C11 uncertainty. Every input has a range. The case must hold at the bad end of the range, not the middle.
- C12 source claims. Speaker forecasts carry the label [SPEAKER CLAIM, EVIDENCE PENDING]. Never build a business case on an unlabeled claim.

U02 , silicon, power, and data centers

- C01 GPU economy. GPUs are the scarcest input. Rent = purchase price divided by lifetime tokens plus margin.
- C02 custom accelerators. Custom silicon beats general GPUs only at sustained high utilization: the crossover is utilization x volume.
- C03 memory/interconnect bottlenecks. FLOPs are cheap, moving data is expensive. Roofline: throughput = min(compute, bandwidth x intensity).
- C04 infrastructure lifecycle. Hardware lives 3-5 years. TCO = buy + power + cooling + refresh.
- C05 power constraints. Power, not silicon, caps the data center. Tokens per day = GPUs x tok/s x U x 86,400.
- C06 cooling. Every watt of compute needs ~0.3 watts of cooling. PUE = total power / IT power.
- C07 gigawatt scale. A gigawatt campus serves ~20M concurrent users at 50 W each. Power is the new moat.
- C08 build/lease tradeoffs. Build for steady base load, lease for peaks. The crossover is utilization x duration.
- C09 utilization. An idle GPU earns nothing. Effective cost = sticker cost / utilization.
- C10 depreciation. Hardware loses ~40 percent of value per year. Book value = purchase x 0.6^t .
- C11 supply concentration. One foundry, one GPU vendor, few grids. Concentration = pricing power upstream.
- C12 scenario analysis. Run base, good, bad. The decision must survive the bad case.

U03 , enterprise infrastructure and token markets

- C01 service-as-software thesis. Software margins with service costs. The thesis holds only if delivery cost per token keeps falling.
- C02 enterprise integration. Integration cost is 3-10x the license. Budget the integration, not the subscription.
- C03 open/closed stacks. Open stacks trade control for cost. Closed stacks trade cost for speed.
- C04 compute supply. Supply = GPUs x utilization x time. Shocks shift supply left and price up.
- C05 token demand. Demand = users x tokens per user. Demand is elastic: price cuts raise volume.
- C06 capacity bottlenecks. The scarcest input sets output. Find the bottleneck before buying anything.
- C07 inference versus training economics. Training is capex-like, inference is opex-like. Unit cost = $C_{\text{train}}/V + c_{\text{inf}}$.
- C08 long contracts. A contract is insurance against price spikes. Regret = contract PV minus market PV.
- C09 ownership. Owning the stack keeps the margin. Renting keeps the optionality. Own the bottleneck.
- C10 data governance. Data has owners, lineage, and retention rules. Governance cost is real and rising.
- C11 unit prices. Quote everything per 1M tokens. The unit price is the only comparable number.
- C12 sensitivity. Vary each input 20 percent. The inputs that move the verdict get measured first.

U04 , enterprise knowledge and inference cloud

- C01 knowledge access value. $\text{Value} = q \times (a_1 - a_0) \times$

v. Toy: $1,000 \times 0.37 \times 20 = 7,400$ per month.

- C02 integration costs. Connectors cost 40k to build,

5k per year to maintain, plus a 30k one-time permission filter. Toy: three sources (20k, 35k, 80k docs), two-year bill = 180k, payback in month 25.

- C03 experience/data feedback. Usage corrects the index, corrections raise quality, quality raises usage. Toy 6-month decayed gain = 9.371 (naive 12.0).

- C04 quality value. Each quality point is worth

w dollars. $\text{Value} = w \times \text{points} \times \text{volume}$.

- C05 hosting/API break-even. $Q^* = F / (p_a - p_h)$.

Toy: $30,000 / 2.50 = 12,000 = 12\text{B}$ tokens per month.

- C06 cache economics. $\text{Effective} = h \times c_c + (1 - h) \times$

c_m . Toy: $0.6 \times 0.25 + 0.4 \times 4.00 = 1.75$ per 1M. Replication (capstone 1) measured H1 FAIL at small caches: analytic models overstate h by ~12 percent.

- C07 latency/quality tradeoff. $\text{Net} = V \times s \times dL -$

saving. Toy: $2\text{M} \times 0.01 \times 0.8 - 30\text{k} = 130\text{k}$ loss, do not batch.

- C08 review labor. $\text{Labor} = \text{tasks} \times e \times \text{minutes}/60 \times$

wage. Toy: $10\text{k} \times 0.10 \times 10/60 \times 40 = 6,666.67$ per month.

- C09 cost per success. $(r+1) \times c / \text{success}$. Toy: $2 \times$

$0.50 / 0.84 = 1.19$ with one retry at $s = 0.6$.

- C10 enterprise pricing. Price the seat for access, the token for usage. Blended margin must stay positive.

- C11 switching costs. Migration cost amortized over the contract. $\text{Payback} = \text{switching} / \text{monthly saving}$.

- C12 inference ROI. $\text{Net} = \text{savings PV} - \text{TCO}$. Toy:

$767.9\text{k} - 680\text{k} = 87.9\text{k}$, pilot the lean design.

U05 , coding/application and life-science economics

- C01 verify gap. Verification is cheaper than generation. The gap is the business: $v - c > 0$.
- C02 software 2.0 economics. Code is written by models, reviewed by humans. Cost = generation + review.
- C03 distribution cost. Serving cost per user falls with scale. Margin = price - $c_{\text{serve}}(\text{user})$.
- C04 build/buy/license. License wins when the vendor amortizes build over many buyers. Compare 3-year TCOs, not year 1.
- C05 Amdahl for AI. Speedup is capped by the non-automatable share: $1 / ((1-p) + p/s)$.
- C06 discovery funnel. Total = $\sum n_i \times c_i$. Toy: 111.002M from discovery to clinical.
- C07 clinical trials. Trials dominate cost: the funnel's tail wags the dog. Price the tail first.
- C08 assay economics. Cost per hit = assays x cost / (assays x survival). Halving survival doubles cost per hit.
- C09 expert review. Experts are the bottleneck: 320 hours per month at 2 experts. Triage multiplies expert hours.
- C10 adoption PV. Gate savings by the ramp: [0.20, 0.50, 0.80] at $r = 10$ percent gives 767.9k, not 1.93M naive.
- C11 life-science quality bar. Accuracy ≥ 0.90 and full audit trails. The bar is regulatory, not technical.
- C12 OOD limits. Out-of-distribution inputs break silently. Monitor the input distribution, not just the output.

U06 , capstone evidence and stakeholder defense

- C01 baseline. Measure the current cost before proposing anything. No baseline, no business case.
- C02 objectives. One primary metric, guardrails for the rest. Pre-register all of them.
- C03 constraints. Budget, latency, compliance: the box the design must fit in.
- C04 quality gates. $a1 \geq 0.75$ or the system does not ship. Gates are binary.
- C05 trust boundaries. Data crosses boundaries only under contract. Mask PII before egress.
- C06 alternatives. Always price the status quo and the hire-more-humans option. The AI design must beat both.
- C07 TCO reckoning. $TCO = B + 3(R + L) + K$. Toy: 1,850k, build share 21.6 percent. Build-only cases understate 3-4x.
- C08 acceptance gates. Cost ≤ 30 , accuracy ≥ 0.90 , escalation ≤ 0.10 , 90 percent confidence. Kill on two consecutive breaches.
- C09 monitoring. Dashboard the five vitals: cost, accuracy, escalation, latency, hit rate. Alert on drift.
- C10 cost/latency assumptions. 1.75 per 1M cached, 0.67 review labor, p99 1.8 s. Mark each for pilot measurement.
- C11 rollout. Shadow, 1 percent, 10 percent, 100 percent. 15-minute rollback, game-day tested.
- C12 falsifiable investment. $L > Y$ on the pilot interval or reject. Toy: [13, 23] vs $Y = 15$: reject. Never move Y or D after seeing data.

The ten numbers to memorize

1. $Q^* = F / (p_a - p_h)$: 12B tokens per month.
 2. Effective cached price: $h \times c_c + (1 - h) \times c_m$.
 3. Cost per success: $(r + 1) \times c / (1 - (1 - s)^{(r+1)})$.
 4. Adoption-gated PV: ramp each year, then discount.
 5. $TCO = B + 3(R + L) + K$: build is ~22 percent.
 6. Break-even volume: $30k / (p_a - p_h)$.
 7. Review labor: $\text{tasks} \times e \times \text{min} / 60 \times \text{wage}$.
 8. Flywheel: decay each month's corrections by $0.9^{\text{(remaining)}}$.
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1. Falsifiable rule: invest iff $L > Y$, pre-registered.
 2. Stakeholder line: measured X, costs Y, saves Z, risk W, pilot 4 weeks, kill criteria named.